全球发明大会中国区
INVENTION CONVENTION CHINA

查新报告

项目名称： Bionic Ankle

团队名称： Justin Guo

团队成员： Justin Guo

查新完成日期： 6 / 23 / 2024

二○二四年

填写说明

一、查新报告格式说明

本报告每栏的大小可随内容需要进行调整，团队成员以手写形式签署查新声明后进行扫描，以 PDF 文件格式上传到 ICC 竞赛系统。

二、查新点

查新点：是指查新项目中的新颖性和创新性之处。

查新要求：（1）是否已经存在与所研究项目相同或相似的研究，涵盖了查找现有文献、专利、技术报告等相关资料，以确保对已有研究工作的充分了解；（2）对所查找到的相关研究项目与查新项目进行深入的对比分析， 重点突出项目之间的异同。包括但不限于研究目的、方法、结果和创新点等方面的比较，以全面评估研究的新颖性和创新性；（3）对查新项目的新颖性和创新性进行明确判断，这需要结合对比分析的结果，清晰地判断项目在当前研究领域中的新颖性和创新性，为项目的价值和意义提供充分的论证和支持。

三、文献检索范围及检索策略

请选择报告中例举的或填写实际查新工作中所使用的文献检索范围（可多选），并且列出检索词等。

四、检索结果

检索结果应当客观准确地反映出对所检数据库和工具书中的相关文献情况进行对比分析的情况。具体来说，检索结果包括：对所检数据库和工具书中的相关文献情况的简要描述；对所列主要相关文献的简要描述，可以使用原文中的摘要或抽取主要信息。

五、查新结论

查新结论应当准确、清晰地反映查新项目的真实情况，以客观、公正的态度呈现。
结论应包括以下内容：
1. 相关文献检出情况：
检索结果涵盖多篇与项目关键内容相关的文献，以不同研究视角进行呈现；
2. 检索结果与查新项目要点的比较分析：
对现有研究与查新项目进行细致比较，凸显出项目的新颖性和创新性之处；
3. 对查新项目新颖性的判断结论：
结合相关文献分析，明确查新项目的新颖性和创新性，强调其独特价值。

六、附件

附件主要包括相关文献的题目、出处以及原文节录的内容。

七、备注

若有其他与项目相关需说明的情况，可进行备注。

八、签名

由项目团队成员、指导教师手写签名。

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项目名称	Bionic Ankle

一．查新项目概况

（查新项目的主要特征、新颖性和创新性、应用范围等）

1. A bionic ankle that can help disabled people to walk
2. Consists of two motors in the upper part of the leg to drive the bionic ankle, a universal joint to act as the human joint, and a movable foot.
3. Used a better spring system to mimic human's ankle function
4. Uses 2 motors to allow a 2-dimensional movement for better adaption
5. Added inversion and eversion movements
6. Used ADAMS to model the kinematics for optimized lengths
7. Uses PID controller to move the leg

二．查新点

（需要查证的能够体现查新项目新颖性和创新性的内容要点、创新点。查新点的表述要客观科学，文字要精练明确，条理清楚。查新项目有多个查新点需要查证时，要逐条列出。）

1. Inversion and Eversion movements
2. Spring system
3. PID controller
4. ADAMS simulation

三．文献检索范围及检索策略

文献检索范围：

□中国知网

□维普期刊网

□万方数据知识服务平台

□中国学术期刊网络出版总库

□国家知识产权局网

□中国国家数字图书馆

□中国重要报纸全文数据库

□百度学术

√Google 学术

□搜索引擎（如百度、谷歌）

□其他：______________________

检索词：

Inversion Eversion bionic ankle

PID bionic ankle

四．检索结果

（依据项目查新点对检出文献进行筛选 ，并对筛选出的中外文密切相关文献和一般相关文献按其与查新点的相关程度及先国内，后国外的顺序分别列出题录，并依据相关文献摘要逐篇进行简要描述。如果查新项目有多个查新点 ， 可逐点分别列出。）

[1] Title: A Multiaxial Bionic Ankle Based on Series Elastic Actuation With a Parallel Spring
Authors: Shun Zhao, Wei Liang, Kunyang Wang, Lei Ren, Zhihui Qian, Guangrong Chen, Xuewei Lu, Di Zhao, Xu Wang, Luquan Ren
Publisher: IEEE

The paper describes the developing of a 2 degree-of-freedom bionic ankle using springs and a four-SEAPS-based spring mechanism. The bionic ankle exhibits a high land-buffering performance.
This paper is similar to my bionic ankle in terms of having a 2 degree-of-freedom, and also a strong parallel spring mechanism.
This paper is different from my bionic ankle in the method to optimize the numbers of the ankle. I used ADAMS to get a virtual environment, which eventually reaches to completely different statistics in the maximum degree to move the ankle, the height of the ankle.

[2] Title: Control of Bionic Robot Leg Performance with Proportional Integral and Derivative Controller
Authors: W. Widhiada, T. G. T. Nindhia, I. M. Widiyarta, and I. N. Budiarsa
Publisher: International Journal of Mechanical Engineering and Robotics Research

The paper introduces the concept of PID controlling to bionic ankles and knees which proves to quickly achieve a stable response.
This paper is similar to my bionic ankle in terms of using a PID controller to quickly, and accurately control the bionic ankle.
My bionic ankle adds a second degree-of-freedom to the human ankle, mimicking the human ankle better.

五．查新结论

（查新结论必须客观、公正、准确、清晰地反映查新项目的真实情况，应包含以下 基本内容：项目查新点归纳、项目查新点与相关文献的逐点对比，对查新项目有 无新颖性和创新性的判断结论。）
After searching on Google Scholar, and comparing the articles, the conclusions are as following:

1. There is an article with a similar mechanical structure, but have different statistics, ways of improvement and approaching the bionic ankle.
2. There is an article with a similar controlling method, but a more limited mobility of the ankle.

There are many similar researches on bionic ankle, the characteristic of my bionic leg are as following:

1. Uses PID controlling mechanism to provide an accurate, quick method to move the ankle to its specific angle.
2. Have inversion and eversion to improve the balancing and adaption of the bionic ankle
3. Using ADAMS to simulate the movement of the bionic ankle for improvements

Some bionic ankles that have been published have some of these characteristics, but do not exhibit them all.

According to the above, the bionic ankle shows innovation through using ADAMS to help create a unique PID controlled 2 DoF bionic ankle.

六．附件清单

（附件主要包括密切相关文献的题目、出处以及原文复制内容。将检索到的全部附件详细内容粘贴至此处））

A Multiaxial Bionic Ankle Based on Series Elastic Actuation With a Parallel Spring

Publisher: IEEE | Cite This | PDF

Shun Zhao ; Wei Liang ; Kunyang Wang ; Lei Ren ; Zhihui Qian ; Guangrong Chen ; Xuewei Lu ;... All Authors

Abstract:
Traditional robotic foot and ankle have received considerable attention for adaptive locomotion on complex terrain and land buffering ability. This study aims to develop a robotic ankle based on series elastic actuator with a parallel spring (SEAPS) by mimicking the two degree-of-freedom human ankle and related muscles. A unique four-SEAPS-based spring mechanism enhances motion adaptation and landing buffer. A dynamic model based on the SEAPS drive is presented, whereas a kinematic solution is realized. The bionic ankle has a wide range of motions with 30° in both the sagittal and frontal planes, which cover most of the human ankle motions. Four experiments were conducted to thoroughly characterize the capabilities. First, the static stiffness and the 2-D/3-D trajectory tracking performance of the proposed ankle were tested. Second, the angle sensing capacity under inclined road surface and the land buffering performance from a specific height were evaluated. The results show that the maximum 3-D motion tracking error is smaller than 2.3%, and the minimum sensing error of inclined road is smaller than 0.5%. The proposed ankle can closely track the 3-D walking trajectory of natural ankle joint with relative root-mean-square error well below 1.0%. Compared with no springs, landing buffer of the ankle

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Traditional robotic foot and ankle have received considerable attention for adaptive locomotion on complex terrain and land buffering ability. This study aims to develop a robotic ankle based on series elastic actuator with a parallel spring (SEAPS) by mimicking the two degree-of-freedom human ankle and related muscles. A unique four-SEAPS-based spring mechanism enhances motion adaptation and landing buffer. A dynamic model based on the SEAPS drive is presented, whereas a kinematic solution is realized. The bionic ankle has a wide range of motions with 30° in both the sagittal and frontal planes, which cover most of the human ankle motions. Four experiments were conducted to thoroughly characterize the capabilities. First, the static stiffness and the 2-D/3-D trajectory tracking performance of the proposed ankle were tested. Second, the angle sensing capacity under inclined road surface and the land buffering performance from a specific height were evaluated. The results show that the maximum 3-D motion tracking error is smaller than 2.3%, and the minimum sensing error of inclined road is smaller than 0.5%. The proposed ankle can closely track the 3-D walking trajectory of natural ankle joint with relative root-mean-square error well below 1.0%. Compared with no springs, landing buffer of the ankle with SEAPS can yield remarkable reduction in both peak ground reaction force (52.2%) and joint torque (57.9%). These findings prove that the SEAPS-based bioinspired robotic ankle exhibits high land buffering performance in an unstructured environment, and also well reproduces the human-like movement in terms of complex 3-D ankle motions.

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Control of Bionic Robot Leg Performance with Proportional Integral and Derivative Controller

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***Abstract*— False limbs are aids for people with disabilities who have not been able to make users move freely like normal people. In this study, the concept of PID control prosthetic limbs was developed with a robot function that resembles the function of the knee ankle joints. By using PID control that the motion of the robot leg can be controlled accurately. The PID parameters obtained for the knee are kp = 2, ki = 18 and kd = 0.5, while for the ankle that is kp = 1, ki = 1.5 and kd = 0.5. The performance of motion bionic robot leg is obtained with small overshoot signal, small error signal and quickly to achieve stable response are 3,432%, 3.69% and 0.877s respectively.**

variables obtained [6]. According Hualong Xie, the bionic robot leg with PID controller control system provides natural human motions using trial and error methods[7].

In this study the auto PID control system on MATLAB / Simulink-based is designed to get an optimal, accurate, and fast response to the prototype bionic foot by reducing the error signal, maximum overshoot, and completion time [8].

II. Method of Research

七．备注

（其他需要注解或说明的内容。例如本项目已经申请过相关领域专利等。）

八．发明团队成员、指导教师签字的查新声明

（签署此查新声明，即表明所提交的查新报告中的陈述事实真实准确，团队成员将对报告中的所有内容负责。）

发明团队成员签名（手写）：

Justin G

发明团队指导教师签名（手写）：